SQL Basics & SELECT

Query Any Database With Confidence

■ LEARNING OUTCOME

By the end of this module you will understand relational databases, write correct SELECT statements, use aliases, sort results with ORDER BY, limit rows, and know the difference between DDL, DML, and DQL commands.

■ Skills You Will Gain

- Understand tables, rows, columns, and primary keys
- Write SELECT statements to query specific columns
- Use DISTINCT to eliminate duplicates
- Sort results with ORDER BY ASC and DESC
- Limit rows with LIMIT and OFFSET for pagination
- Use column and table aliases for readable queries

■ **SQL is the most widely used data language on Earth. Every database -- PostgreSQL, MySQL, SQLite, SQL Server, BigQuery, Snowflake, Redshift -- uses SQL. One skill, infinite databases.**

SECTION 1

The Relational Database Model

Core Concepts

TABLE: A grid of data -- like a spreadsheet tab. Rows = records, Columns = fields. ROW: One record (one person, one order, one product). COLUMN: One attribute of every record (name, price, date). PRIMARY KEY: Unique identifier for each row -- no nulls, no duplicates. FOREIGN KEY: Column in one table that references a primary key in another table. SCHEMA: The structure definition -- which tables exist, which columns, which types. SQL Command Categories: DDL (Data Definition): CREATE TABLE, ALTER TABLE, DROP TABLE DML (Data Manipulation): INSERT, UPDATE, DELETE DQL (Data Query): SELECT -- the most important for analysts DCL (Data Control): GRANT, REVOKE TCL (Transaction Control):COMMIT, ROLLBACK, SAVEPOINT

SECTION 2

Data Types Reference

Category	Data Type	Description	Example
Text	VARCHAR(n)	Variable-length string up to n chars	VARCHAR(100) for names
Text	CHAR(n)	Fixed-length string -- pads with spaces	CHAR(2) for country code
Text	TEXT	Unlimited length text	Blog posts, descriptions
Number	INT / INTEGER	Whole numbers (-2B to 2B)	age, quantity, count
Number	BIGINT	Large whole numbers	user_id at massive scale
Number	DECIMAL(p,s)	Exact decimal -- p digits, s after point	DECIMAL(10,2) for price
Number	FLOAT/DOUBLE	Approximate decimal -- scientific	Measurements, coordinates
Date	DATE	Date only YYYY-MM-DD	2024-01-15
Date	TIMESTAMP	Date + time YYYY-MM-DD HH:MM:SS	2024-01-15 10:30:00
Other	BOOLEAN	True or false	is_active, is_premium
Other	NULL	Absence of value -- not zero, not empty	Unknown phone number

SECTION 3

CREATE TABLE and INSERT

```
-- Create a customers table CREATE TABLE customers ( customer_id INT PRIMARY KEY, first_name VARCHAR(50) NOT NULL, last_name VARCHAR(50) NOT NULL, email VARCHAR(100) UNIQUE NOT NULL, phone VARCHAR(20), city VARCHAR(50) DEFAULT 'Unknown', signup_date DATE NOT NULL, is_premium BOOLEAN DEFAULT FALSE, annual_spend DECIMAL(12,2) DEFAULT 0.00 ); -- Create orders table with foreign key CREATE TABLE orders ( order_id INT PRIMARY KEY, customer_id INT NOT NULL, order_date DATE NOT NULL, total_amount DECIMAL(12,2) NOT NULL, status VARCHAR(20) DEFAULT 'pending', FOREIGN KEY (customer_id) REFERENCES customers(customer_id) ); -- INSERT rows INSERT INTO customers (customer_id, first_name, last_name, email, city, signup_date) VALUES (1, 'Alice', 'Sharma', 'alice@example.com', 'Mumbai', '2022-03-15'), (2, 'Bob', 'Patel', 'bob@example.com', 'Delhi', '2021-07-20'), (3, 'Carol', 'Singh', 'carol@example.com', 'Bangalore', '2023-01-10'), (4, 'David', 'Kumar', 'david@example.com', 'Chennai', '2022-11-05');
```

SECTION 4

SELECT -- The Most Important Statement

```
-- Basic SELECT: choose specific columns SELECT first_name, last_name, email FROM customers; -- SELECT ALL columns (avoid in production -- slow, fragile) SELECT * FROM customers; -- Column ALIASES -- rename output columns SELECT first_name AS first, last_name AS last, annual_spend AS total_spent, annual_spend * 0.18 AS gst_amount, CONCAT(first_name,' ',last_name) AS full_name FROM customers; -- Table ALIAS -- shorter table reference name SELECT c.first_name, c.city, c.annual_spend FROM customers AS c; -- or just: customers c -- DISTINCT -- remove duplicate values SELECT DISTINCT city FROM customers; -- unique cities SELECT DISTINCT status FROM orders; -- unique order statuses SELECT DISTINCT city, is_premium -- distinct combinations FROM customers; -- ORDER BY -- sort results SELECT * FROM customers ORDER BY last_name ASC; -- alphabetical SELECT * FROM orders ORDER BY total_amount DESC; -- highest first SELECT * FROM customers ORDER BY city ASC, annual_spend DESC; -- multi-column sort -- LIMIT and OFFSET -- pagination SELECT * FROM customers LIMIT 10; -- first 10 rows SELECT * FROM customers LIMIT 10 OFFSET 20; -- rows 21-30 (page 3) SELECT * FROM products ORDER BY price DESC LIMIT 5; -- top 5 most expensive -- Expressions in SELECT SELECT product_name, price, price * 1.18 AS price_with_gst, price * 0.85 AS discounted_price, ROUND(price / 83.5, 2) AS price_usd, UPPER(product_name) AS name_upper, LENGTH(product_name) AS name_length FROM products;
```

SECTION 5

SQL Execution Order

The Order SQL Clauses Execute (NOT the order you write them)

1. FROM -- which table(s) to use 2. JOIN -- combine tables 3. WHERE -- filter individual rows 4. GROUP BY -- group remaining rows 5. HAVING -- filter groups 6. SELECT -- choose columns (aliases defined here) 7. DISTINCT -- remove duplicates 8. ORDER BY -- sort (can use SELECT aliases) 9. LIMIT -- cut rows This explains why you cannot use SELECT aliases in WHERE clauses -- WHERE executes before SELECT, so the aliases don't exist yet!

■ PRACTICE Q & A

Q1. What is the difference between a PRIMARY KEY and a FOREIGN KEY?

> ***A:*** *PRIMARY KEY: unique identifier for each row in a table -- no nulls, no duplicates. FOREIGN KEY: a column in one table that references the PRIMARY KEY of another table -- enforces referential integrity between tables.*

Q2. Why should you avoid SELECT * in production queries?

> ***A:*** *SELECT * fetches all columns even ones you don't need -- wastes network bandwidth and memory. If columns are added/reordered later, application code breaks. Always specify the exact columns you need.*

Q3. What is the difference between LIMIT and OFFSET?

> ***A:*** *LIMIT caps the number of rows returned. OFFSET skips a number of rows before starting. LIMIT 10 OFFSET 20 skips 20 rows and returns the next 10 -- this is page 3 of 10-item pages.*

Q4. What is the difference between NULL and an empty string?

***A:** NULL means the value is unknown or missing -- no data at all. Empty string '' is a known value -- it is a string with zero characters. NULL != NULL in SQL (use IS NULL to check, not = NULL).*

Q5. What is the execution order of SQL clauses?

***A:** FROM -> JOIN -> WHERE -> GROUP BY -> HAVING -> SELECT -> DISTINCT -> ORDER BY -> LIMIT. This is why column aliases defined in SELECT cannot be used in WHERE -- WHERE runs first.*

■■ SQL Basics Cheat Sheet	
`SELECT col1, col2 FROM tbl`	Select specific columns
`SELECT * FROM tbl`	All columns -- avoid in production
`SELECT DISTINCT col FROM tbl`	Unique values only
`col AS alias`	Rename column in output
`table AS t`	Short table alias
`ORDER BY col ASC/DESC`	Sort results
`ORDER BY col1 ASC, col2 DESC`	Multi-column sort
`LIMIT 10`	First 10 rows
`LIMIT 10 OFFSET 20`	Rows 21-30 (pagination)
`VARCHAR(100)`	Variable-length text
`DECIMAL(10,2)`	Exact decimal for money
`NOT NULL / UNIQUE / DEFAULT`	Column constraints